

INTEGER SOLUTIONS OF $z^2 + y^2z + x^3 = 2$

ABSTRACT. We give an exact divisor-theoretic classification of every integer triple (x, y, z) satisfying $z^2 + y^2z + x^3 = 2$. For $x \geq 2$, solutions correspond canonically to positive factor pairs of $x^3 - 2$ whose sum is a square; for $x \leq 1$, they correspond canonically to positive factor pairs of $2 - x^3$ whose difference is a square. The correspondence yields exact formulas for the number of solutions above each fixed value of x . We also prove the natural symmetry, parity, congruence, coprimality, sign, and size restrictions satisfied by every solution.

1. THE COMPLETE CLASSIFICATION

We study the equation

$$(1) \quad z^2 + y^2z + x^3 - 2 = 0, \quad (x, y, z) \in \mathbb{Z}^3.$$

Its decisive form is

$$(2) \quad z(z + y^2) = 2 - x^3.$$

Since x is an integer, exactly one of $x \leq 1$ and $x \geq 2$ holds. These two ranges give opposite signs on the right-hand side of (2), and this sign change accounts for the two forms in the classification below.

Theorem 1.1 (Complete classification). *All integer solutions of (1) are given by the following two disjoint constructions.*

(i) The range $x \geq 2$. Choose integers

$$x \geq 2, \quad 1 \leq d < e, \quad m \geq 1$$

such that

$$(3) \quad de = x^3 - 2, \quad d + e = m^2.$$

Then the four triples

$$(4) \quad (x, m, -d), \quad (x, -m, -d), \quad (x, m, -e), \quad (x, -m, -e)$$

are solutions. Conversely, every solution with $x \geq 2$ occurs in exactly one such four-element family.

(ii) The range $x \leq 1$. Choose integers

$$x \leq 1, \quad 1 \leq d \leq e, \quad m \geq 0$$

such that

$$(5) \quad de = 2 - x^3, \quad e - d = m^2.$$

Then

$$(6) \quad (x, \varepsilon m, d), \quad (x, \varepsilon m, -e), \quad \varepsilon \in \{1, -1\},$$

are solutions, with repetitions discarded when $m = 0$. Conversely, every solution with $x \leq 1$ occurs in exactly one family arising from an ordered factor pair $d \leq e$ satisfying (5). Such a family has four elements when $d < e$ and two elements when $d = e$.

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Proof. We prove both implications in each range and then verify the asserted uniqueness.

Suppose first that $x \geq 2$, and set

$$N = x^3 - 2 > 0.$$

Equation (2) becomes

$$(7) \quad z(z + y^2) = -N < 0.$$

Thus z and $z + y^2$ have opposite signs. Since $z + y^2 \geq z$, the only possible ordering is

$$(8) \quad z < 0 < z + y^2.$$

Define

$$a = -z, \quad b = z + y^2.$$

Then $a, b \in \mathbb{Z}_{>0}$, and (7) gives

$$(9) \quad ab = (-z)(z + y^2) = N = x^3 - 2.$$

Moreover,

$$(10) \quad a + b = -z + (z + y^2) = y^2.$$

Let

$$d = \min\{a, b\}, \quad e = \max\{a, b\}, \quad m = |y|.$$

Then $de = x^3 - 2$ and $d + e = m^2$. It remains to prove that $d \neq e$. If $d = e$, then (10) gives $m^2 = 2d$. Hence m is even, say $m = 2r$, and then $d = 2r^2$. By (9),

$$x^3 = d^2 + 2 = 4r^4 + 2 \equiv 2 \pmod{4}.$$

This is impossible, because an integer cube modulo 4 is congruent only to 0, 1, or 3. Therefore $d < e$. Since a is either d or e , the equality $z = -a$ shows that the original solution is one of the four triples in (4).

Conversely, assume (3). If $y = \varepsilon m$ and $z = -d$, then

$$\begin{aligned} z^2 + y^2 z + x^3 - 2 &= d^2 - m^2 d + (x^3 - 2) \\ &= d^2 - d(d + e) + de \\ &= 0. \end{aligned}$$

The same calculation with d and e interchanged proves the assertion for $z = -e$. The four triples are distinct because $m > 0$ and $d < e$.

The construction is canonical. Indeed, the solution itself recovers

$$\{d, e\} = \{-z, z + y^2\}, \quad m = |y|,$$

with the two elements placed in increasing order. Thus two different data sets satisfying (3) cannot yield the same solution.

Now suppose that $x \leq 1$, and put

$$M = 2 - x^3 > 0.$$

Equation (2) becomes

$$(11) \quad z(z + y^2) = M > 0.$$

Consequently, z and $z + y^2$ are nonzero and have the same sign.

If both are positive, set

$$d = z, \quad e = z + y^2.$$

Then $1 \leq d \leq e$, and

$$de = M, \quad e - d = y^2.$$

If both are negative, set

$$d = -(z + y^2), \quad e = -z.$$

Since $z + y^2 \geq z$, we again have $1 \leq d \leq e$, and

$$de = (-z - y^2)(-z) = z(z + y^2) = M, \quad e - d = -z + (z + y^2) = y^2.$$

In either case, with $m = |y|$, the data satisfy (5). In the positive-sign case $z = d$, whereas in the negative-sign case $z = -e$, so the solution has one of the forms in (6).

Conversely, suppose (5) holds. Since $x^3 - 2 = -de$, for $y = \varepsilon m$ and $z = d$ we obtain

$$\begin{aligned} z^2 + y^2 z + x^3 - 2 &= d^2 + m^2 d - de \\ &= d^2 + d(e - d) - de \\ &= 0. \end{aligned}$$

For $z = -e$, we similarly obtain

$$\begin{aligned} z^2 + y^2 z + x^3 - 2 &= e^2 - m^2 e - de \\ &= e^2 - e(e - d) - de \\ &= 0. \end{aligned}$$

If $d < e$, then $m^2 = e - d > 0$, so $m > 0$ and the four displayed triples are distinct. If $d = e$, then $m = 0$, and the construction yields precisely the two distinct triples $(x, 0, d)$ and $(x, 0, -d)$.

Finally, the data are again recovered canonically from a solution. When $z > 0$, one has

$$(d, e) = (z, z + y^2),$$

and when $z + y^2 < 0$, one has

$$(d, e) = (-(z + y^2), -z).$$

In both cases $m = |y|$. This proves uniqueness and completes the proof. $\square$

Corollary 1.2 (Divisor tests and exact fiber sizes). *Fix $x \in \mathbb{Z}$, and let S_x denote the set of pairs $(y, z) \in \mathbb{Z}^2$ for which (x, y, z) satisfies (1).*

If $x \geq 2$, put $N = x^3 - 2$ and define

$$\mathcal{D}_+(x) = \left\{ d \in \mathbb{Z}_{>0} : d \mid N, \quad d < \sqrt{N}, \quad d + \frac{N}{d} \text{ is a perfect square} \right\}.$$

Then

$$(12) \quad |S_x| = 4|\mathcal{D}_+(x)|.$$

If $x \leq 1$, put $M = 2 - x^3$ and define

$$\mathcal{D}_-(x) = \left\{ d \in \mathbb{Z}_{>0} : d \mid M, \quad d < \sqrt{M}, \quad \frac{M}{d} - d \text{ is a perfect square} \right\}.$$

Then

$$(13) \quad |S_x| = 4|\mathcal{D}_-(x)| + 2\mathbf{1}_{\{M \text{ is a perfect square}\}}.$$

In particular, S_x is finite for every fixed integer x .

Proof. Assume first that $x \geq 2$. Every positive unordered factor pair of N has a unique representation

$$\left(d, \frac{N}{d} \right), \quad d \mid N, \quad d \leq \sqrt{N}.$$

The equality case $d = \sqrt{N}$ cannot satisfy the square-sum condition occurring in Theorem 1.1(i), because that would give an admissible pair with $d = e$, which was proved impossible. Hence each admissible factor pair is represented by exactly one $d \in \mathcal{D}_+(x)$, and Theorem 1.1(i) associates exactly four distinct solutions to it. Canonical uniqueness shows that the resulting four-element families are disjoint. This proves (12).

Now assume that $x \leq 1$. Every positive factor pair $d \leq e$ of M is represented uniquely by a divisor $d \leq \sqrt{M}$. A pair with $d < e$ contributes four solutions exactly when $e - d = M/d - d$ is a square. The central pair $d = e = \sqrt{M}$ exists exactly when M is a square; its difference is 0, and it contributes exactly the two solutions with $y = 0$. Again, canonical uniqueness makes all contributions disjoint. Formula (13) follows. Since a nonzero integer has only finitely many positive divisors, both fibers are finite. $\square$

Corollary 1.3 (Terminating exact procedure). *For any prescribed integer x , all corresponding pairs (y, z) can be determined by a finite calculation: factor $|x^3 - 2|$, list one representative from each positive factor pair, and apply the square-sum test when $x \geq 2$ or the square-difference test when $x \leq 1$.*

Proof. This is precisely the construction in Theorem 1.1, organized by the unique divisor representatives used in Corollary 1.2. The procedure terminates because $|x^3 - 2|$ is a positive integer and therefore has finitely many positive divisors. $\square$

2. ARITHMETIC CONSEQUENCES

Proposition 2.1 (The two involutions). *The solution set of (1) is invariant under*

$$\sigma(x, y, z) = (x, -y, z)$$

and

$$\tau(x, y, z) = (x, y, -y^2 - z).$$

The maps σ and τ are commuting involutions. A solution with $y \neq 0$ belongs to an orbit of exactly four solutions under the group they generate, whereas a solution with $y = 0$ belongs to an orbit of exactly two solutions.

Proof. The equation depends on y only through y^2 , so it is invariant under σ . Also,

$$(-y^2 - z)((-y^2 - z) + y^2) = (-y^2 - z)(-z) = z(z + y^2),$$

so the factorized equation (2) is invariant under τ . Direct substitution shows

$$\sigma^2 = \tau^2 = \text{id}, \quad \sigma\tau = \tau\sigma.$$

We next prove that τ fixes no integer solution. If $\tau(x, y, z) = (x, y, z)$, then

$$2z = -y^2.$$

Thus y is even; write $y = 2r$, so $z = -2r^2$. Substitution into (1) gives

$$x^3 = 4r^4 + 2 \equiv 2 \pmod{4},$$

which is impossible for an integer cube. Hence τ has no fixed solution.

If $y \neq 0$, then σ has no fixed point, and $\sigma\tau$ cannot fix the solution because its second coordinate is $-y \neq y$. Thus no nonidentity element of the four-element group $\{\text{id}, \sigma, \tau, \sigma\tau\}$ fixes the solution, so its orbit has four elements.

If $y = 0$, then σ fixes the solution. The equation becomes $z^2 + x^3 = 2$, and $z \neq 0$, since $z = 0$ would force the impossible equality $x^3 = 2$. Therefore $\tau(x, 0, z) = (x, 0, -z)$ is distinct from $(x, 0, z)$, and the orbit has exactly two elements. $\square$

Proposition 2.2 (Parity and congruence restrictions). *Every integer solution of (1) satisfies the following assertions.*

- (1) x and y have opposite parity.
- (2) If x is odd, then y is even, z is odd, and $x \equiv 1 \pmod{4}$. More precisely,

$$\begin{cases} x \equiv 1 \pmod{8}, & \text{if } 4 \mid y, \\ x \equiv 5 \pmod{8}, & \text{if } y \equiv 2 \pmod{4}. \end{cases}$$

- (3) If x is even, then y is odd and

$$z \equiv 1 \text{ or } 6 \pmod{8}.$$

Proof. Reducing (1) modulo 2 and using $u^2 \equiv u \pmod{2}$ gives

$$z + yz + x \equiv z(1 + y) + x \equiv 0 \pmod{2}.$$

If y is odd, then $1 + y \equiv 0 \pmod{2}$, so x is even.

Suppose instead that y is even. The same congruence gives $z \equiv x \pmod{2}$. If x were even, then z would also be even. In that event z^2 , y^2z , and x^3 would all be divisible by 4, contradicting the equality

$$z^2 + y^2z + x^3 = 2.$$

Therefore x is odd. This proves that x and y have opposite parity. It also proves that when x is odd, y is even and z is odd.

Assume x is odd. Modulo 4, one has $z^2 \equiv 1$ and $y^2z \equiv 0$, so

$$1 + x^3 \equiv 2 \pmod{4}.$$

Hence $x^3 \equiv 1 \pmod{4}$, and therefore $x \equiv 1 \pmod{4}$.

For the refinement modulo 8, recall that an odd integer satisfies $z^2 \equiv 1 \pmod{8}$ and $x^3 \equiv x \pmod{8}$. If $4 \mid y$, then $y^2z \equiv 0 \pmod{8}$, whence

$$1 + x \equiv 2 \pmod{8}$$

and $x \equiv 1 \pmod{8}$. If $y \equiv 2 \pmod{4}$, then $y^2 \equiv 4 \pmod{8}$; because z is odd, $y^2z \equiv 4 \pmod{8}$. Hence

$$1 + 4 + x \equiv 2 \pmod{8},$$

which gives $x \equiv 5 \pmod{8}$.

Finally, assume x is even. Then y is odd by the first part. Consequently $x^3 \equiv 0 \pmod{8}$ and $y^2 \equiv 1 \pmod{8}$, so (1) gives

$$z^2 + z \equiv 2 \pmod{8}.$$

Checking the eight residue classes modulo 8 shows that this congruence holds exactly for $z \equiv 1$ and $z \equiv 6 \pmod{8}$. $\square$

Proposition 2.3 (Coprimality with the cubic variable). *For every integer solution of (1),*

$$(14) \quad \gcd(x, z) \mid 2, \quad \gcd(x, z + y^2) \mid 2.$$

In particular, if x is odd, then

$$\gcd(x, z) = \gcd(x, z + y^2) = 1.$$

Proof. Let $g = \gcd(x, z)$. Then g divides x^3 and also divides $z(z + y^2)$. By (2),

$$x^3 + z(z + y^2) = 2,$$

so $g \mid 2$.

Likewise, if $h = \gcd(x, z + y^2)$, then h divides x^3 and $z(z + y^2)$, and the same identity gives $h \mid 2$. When x is odd, both greatest common divisors are odd divisors of 2, hence equal to 1. $\square$

Proposition 2.4 (Sign and size restrictions). *Let (x, y, z) be a solution of (1).*

(1) *If $x \geq 2$, then*

$$(15) \quad -y^2 < z < 0$$

and

$$(16) \quad 2\sqrt{x^3 - 2} < y^2 \leq x^3 - 1.$$

(2) *If $x \leq 1$, then exactly one of*

$$(17) \quad z > 0, \quad z < -y^2$$

holds, and

$$(18) \quad 0 \leq y^2 \leq 1 - x^3.$$

Proof. Assume first that $x \geq 2$. The ordering (8) is exactly $-y^2 < z < 0$, proving (15). With the notation of Theorem 1.1(i),

$$y^2 = d + e, \quad de = x^3 - 2, \quad d < e.$$

The strict arithmetic-geometric mean inequality gives

$$d + e > 2\sqrt{de} = 2\sqrt{x^3 - 2}.$$

Also,

$$de + 1 - (d + e) = (d - 1)(e - 1) \geq 0,$$

so

$$y^2 = d + e \leq de + 1 = x^3 - 1.$$

This proves (16).

Now assume that $x \leq 1$. By (11), either both z and $z + y^2$ are positive or both are negative. In the first case $z > 0$. In the second case $z + y^2 < 0$, which is equivalent to $z < -y^2$. The alternatives are mutually exclusive, proving (17).

With the notation of Theorem 1.1(ii),

$$y^2 = e - d, \quad de = 2 - x^3, \quad 1 \leq d \leq e.$$

Since $d \geq 1$, one has $e \leq de = 2 - x^3$. Therefore

$$0 \leq y^2 = e - d \leq e - 1 \leq 1 - x^3,$$

which proves (18). $\square$

Remark 2.5. For fixed x and y , equation (1) is quadratic in z . If one root is z , the other is $-y^2 - z$, exactly as in Proposition 2.1. Theorem 1.1 identifies these two roots as $-d, -e$ when $x \geq 2$, and as $d, -e$ when $x \leq 1$.

3. CERTIFIED EXAMPLES

Each row below records a factor pair and the required square relation. Theorem 1.1 then supplies all corresponding sign choices for y and both values of z .

x	factorization	square relation	resulting (y, z) values
1	$2 - 1^3 = 1 \cdot 1$	$1 - 1 = 0^2$	$(0, 1), (0, -1)$
0	$2 - 0^3 = 1 \cdot 2$	$2 - 1 = 1^2$	$(\pm 1, 1), (\pm 1, -2)$
-2	$2 - (-2)^3 = 1 \cdot 10$	$10 - 1 = 3^2$	$(\pm 3, 1), (\pm 3, -10)$
-7	$2 - (-7)^3 = 5 \cdot 69$	$69 - 5 = 8^2$	$(\pm 8, 5), (\pm 8, -69)$
8	$8^3 - 2 = 15 \cdot 34$	$15 + 34 = 7^2$	$(\pm 7, -15), (\pm 7, -34)$
20	$20^3 - 2 = 31 \cdot 258$	$31 + 258 = 17^2$	$(\pm 17, -31), (\pm 17, -258)$
26	$26^3 - 2 = 87 \cdot 202$	$87 + 202 = 17^2$	$(\pm 17, -87), (\pm 17, -202)$
26	$26^3 - 2 = 58 \cdot 303$	$58 + 303 = 19^2$	$(\pm 19, -58), (\pm 19, -303)$

The same criterion certifies solutions with much larger coordinates. For example,

$$483308^3 - 2 = 84466 \cdot 1336564813335$$

together with

$$84466 + 1336564813335 = 1156099^2.$$

Hence the four triples

$$(483308, \pm 1156099, -84466), \quad (483308, \pm 1156099, -1336564813335)$$

are exact integer solutions.

Remark 3.1 (Direct verification of every constructed triple). No example, and no solution in the classification, rests on numerical approximation. In the range $x \geq 2$, the identities $de = x^3 - 2$ and $d + e = m^2$ give

$$d^2 - m^2d + x^3 - 2 = d^2 - d(d + e) + de = 0$$

and the corresponding identity with d and e interchanged. In the range $x \leq 1$, the identities $de = 2 - x^3$ and $e - d = m^2$ give

$$d^2 + m^2d + x^3 - 2 = d^2 + d(e - d) - de = 0$$

and

$$e^2 - m^2e + x^3 - 2 = e^2 - e(e - d) - de = 0.$$

Thus the factor certificates displayed above verify the associated triples exactly.